

Customer Shopping Behaviour Analysis

Transactional insights from **3,900 purchases** across product categories – uncovering spending patterns, segments, and growth opportunities.



Dataset at a Glance

3,900

Total Purchases

18

Features

37

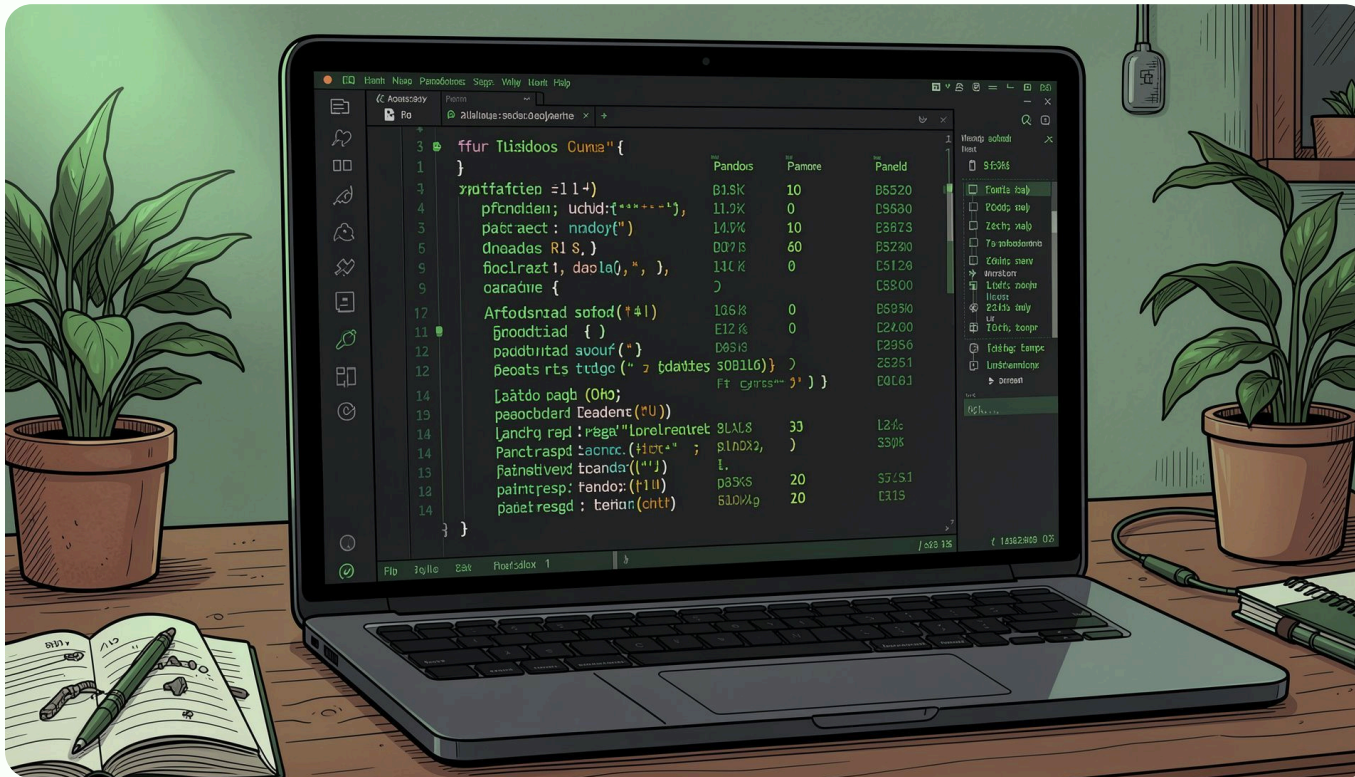
Missing Values

Review Rating column only

50

Locations

Data Preparation in Python



→ **Clean & Standardize**

Imputed missing ratings;
renamed columns to snake_case

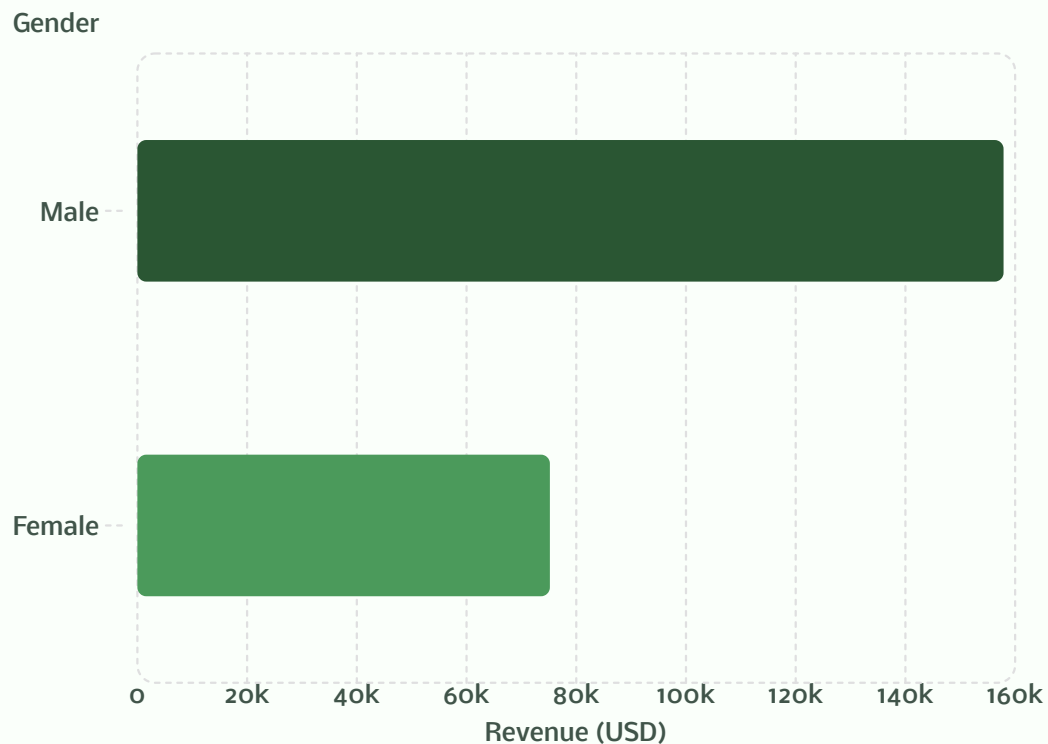
→ **Feature Engineering**

Created age_group &
purchase_frequency_days

→ **Load to PostgreSQL**

Cleaned DataFrame pushed to
database for SQL analysis

Revenue & Spending Insights



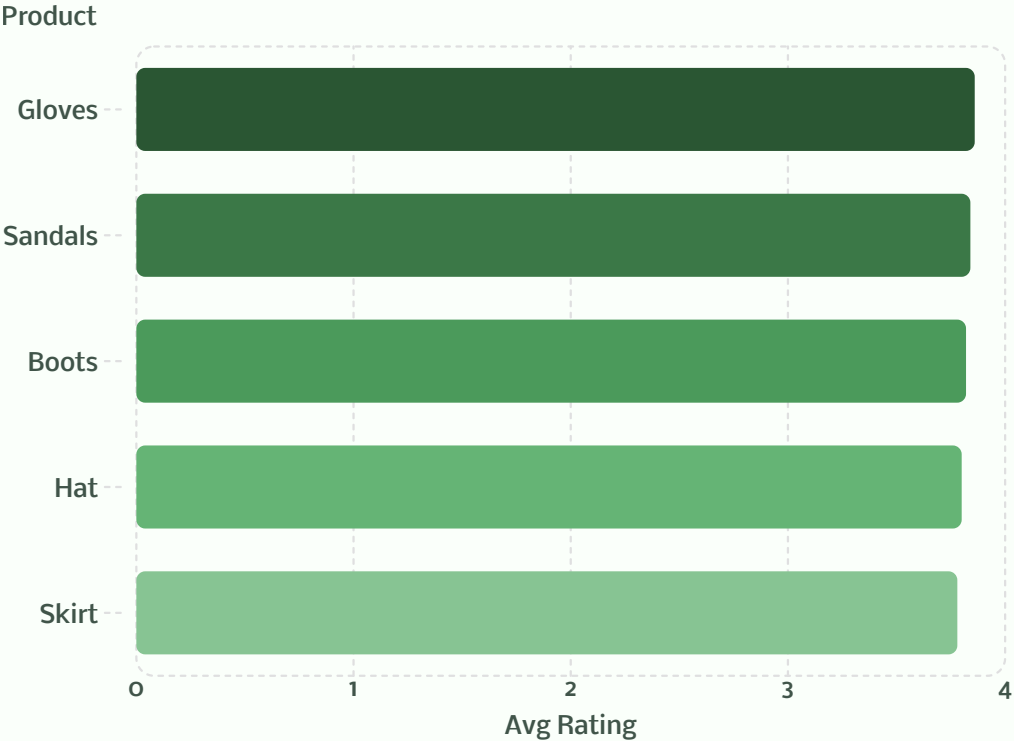
Key Findings

Male customers generate 2× more revenue

839 high-spenders used discounts

Avg purchase: \$59.76

Top-Rated Products



Overall Avg Rating: 3.75 / 5

Footwear and Accessories dominate top ratings – strong candidates for featured campaigns.

Subscription & Shipping Behavior

Subscribers (27%)

1,053 customers · \$59.49 avg spend · **\$62,645** total revenue

Non-Subscribers (73%)

2,847 customers · \$59.87 avg spend · **\$170,436** total revenue

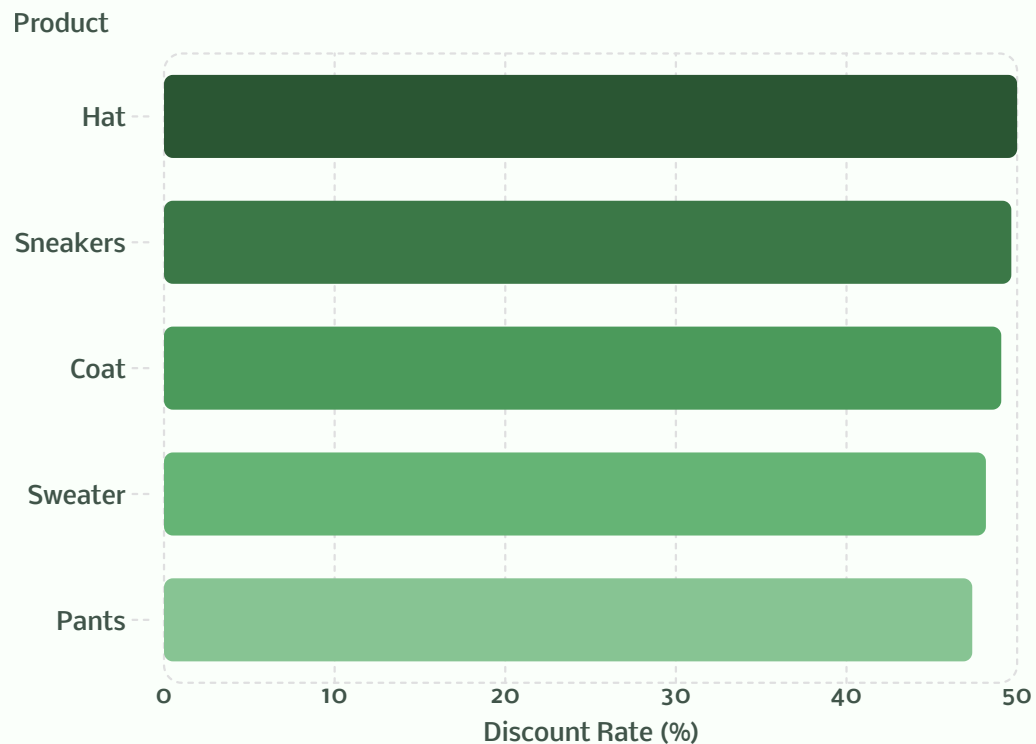
Express Shipping

Avg purchase **\$60.48** vs Standard \$58.46

Repeat Buyers

958 subscribers made >5 purchases vs 2,518 non-subscribers

Discount-Dependent Products



~50% of sales rely on discounts

Hats, Sneakers, and Coats are most discount-sensitive – review margin impact carefully.

Customer Segmentation

 **Loyal**

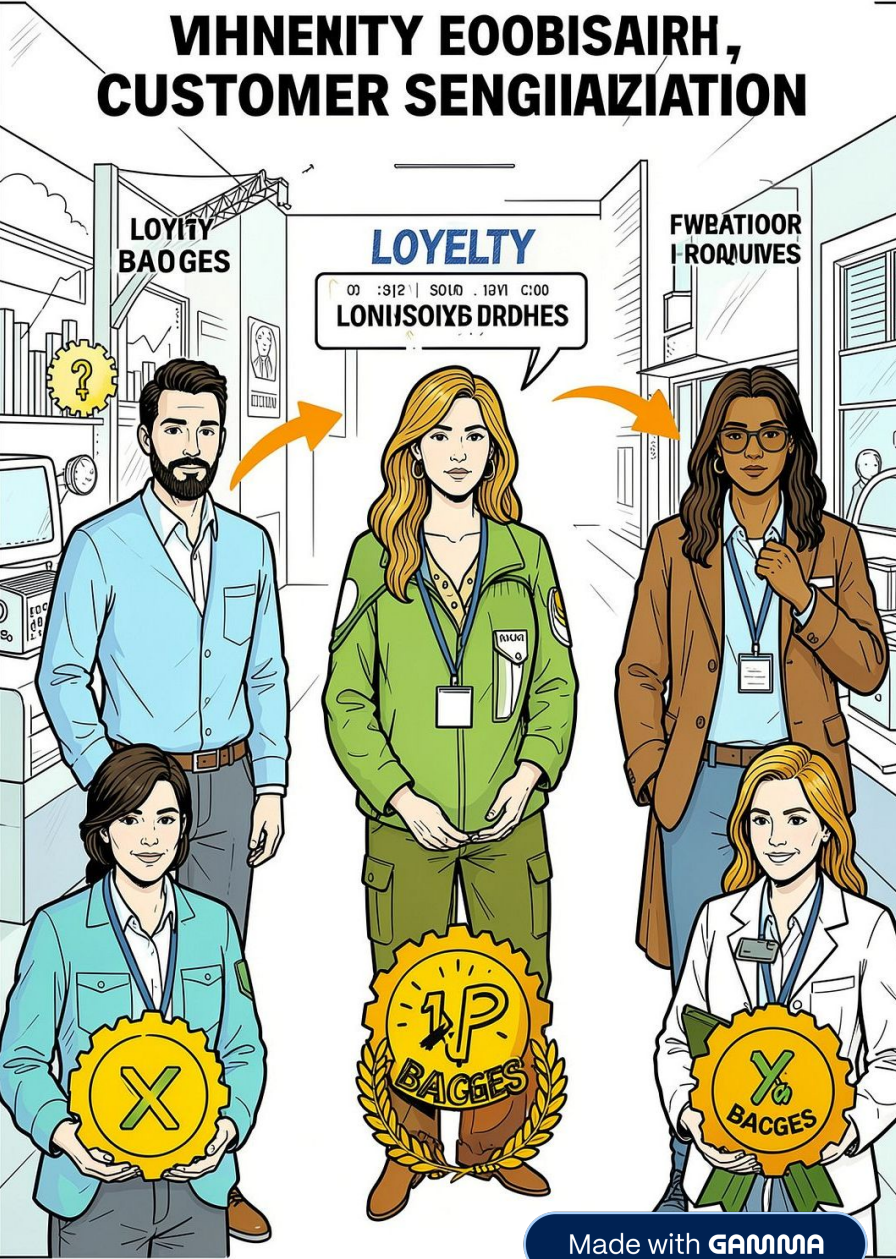
3,116 customers

 **Returning**

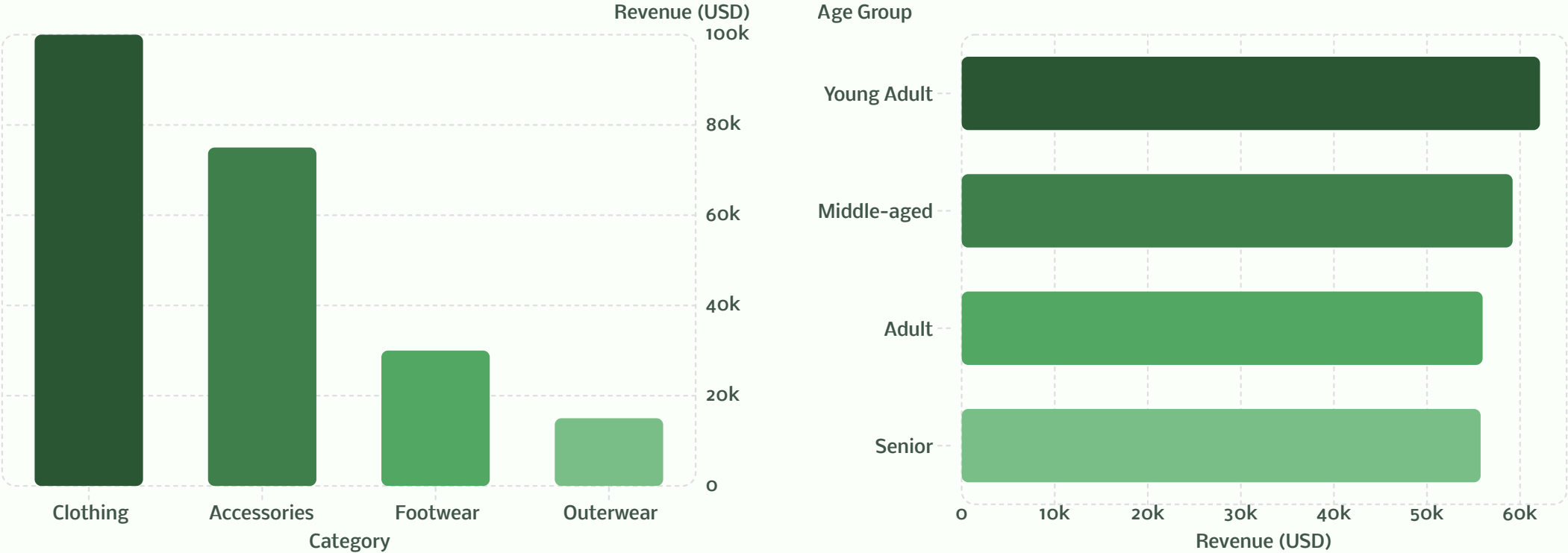
701 customers

 **New**

83 customers



Revenue by Category & Age Group



Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits



Loyalty Programs

Reward repeat buyers toward "Loyal" segment



Review Discount Policy

Balance sales lift with margin control



Targeted Marketing

Focus on high-revenue age groups & express-shipping users

